

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
df = pd.read_csv ("/content/winedataset.csv")
```

```
print(df.head())
```

	Wine	Alcohol	Malic.acid	Ash	Ac1	Mg	Phenols	Flavanoids	\
0	1	14.23	1.71	2.43	15.6	127	2.80	3.06	
1	1	13.20	1.78	2.14	11.2	100	2.65	2.76	
2	1	13.16	2.36	2.67	18.6	101	2.80	3.24	
3	1	14.37	1.95	2.50	16.8	113	3.85	3.49	
4	1	13.24	2.59	2.87	21.0	118	2.80	2.69	

	Nonflavanoid.phenols	Proanth	Color.int	Hue	OD	Proline
0	0.28	2.29	5.64	1.04	3.92	1065
1	0.26	1.28	4.38	1.05	3.40	1050
2	0.30	2.81	5.68	1.03	3.17	1185
3	0.24	2.18	7.80	0.86	3.45	1480
4	0.39	1.82	4.32	1.04	2.93	735

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Wine                                  178 non-null    int64
1   Alcohol                              178 non-null    float64
2   Malic.acid                           178 non-null    float64
3   Ash                                  178 non-null    float64
4   Ac1                                  178 non-null    float64
5   Mg                                   178 non-null    int64
6   Phenols                              178 non-null    float64
7   Flavanoids                           178 non-null    float64
8   Nonflavanoid.phenols                 178 non-null    float64
9   Proanth                              178 non-null    float64
10  Color.int                            178 non-null    float64
11  Hue                                  178 non-null    float64
12  OD                                   178 non-null    float64
13  Proline                              178 non-null    int64
dtypes: float64(11), int64(3)
memory usage: 19.6 KB
None
```

```
print(df.isnull().sum())
```

Wine	0
Alcohol	0
Malic.acid	0
Ash	0
Ac1	0
Mg	0
Phenols	0
Flavanoids	0
Nonflavanoid.phenols	0
Proanth	0
Color.int	0
Hue	0
OD	0

```
Proline          0
dtype: int64
```

```
x = df.drop("Wine", axis=1)
y = df['Wine']
```

```
x_train, x_test, y_train, y_test, = train_test_split(
x, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(x_train, y_train)
```

/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
    LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(x_test)
```

```
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
```

```
Accuracy: 1.0
```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Confusion Matrix:
```

```
[[14  0  0]
 [ 0 14  0]
 [ 0  0  8]]
```

```
print("\nClassification report:")
print(classification_report(y_test, y_pred))
```

```
Classification report:
              precision    recall  f1-score   support

     1             1.00      1.00      1.00        14
     2             1.00      1.00      1.00        14
     3             1.00      1.00      1.00         8

   accuracy                   1.00        36
  macro avg              1.00      1.00      1.00        36
 weighted avg              1.00      1.00      1.00        36
```